

Ethanol Blending in Gasoline – South Korea

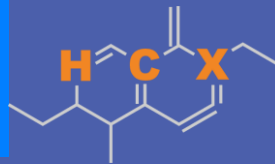
October 2025



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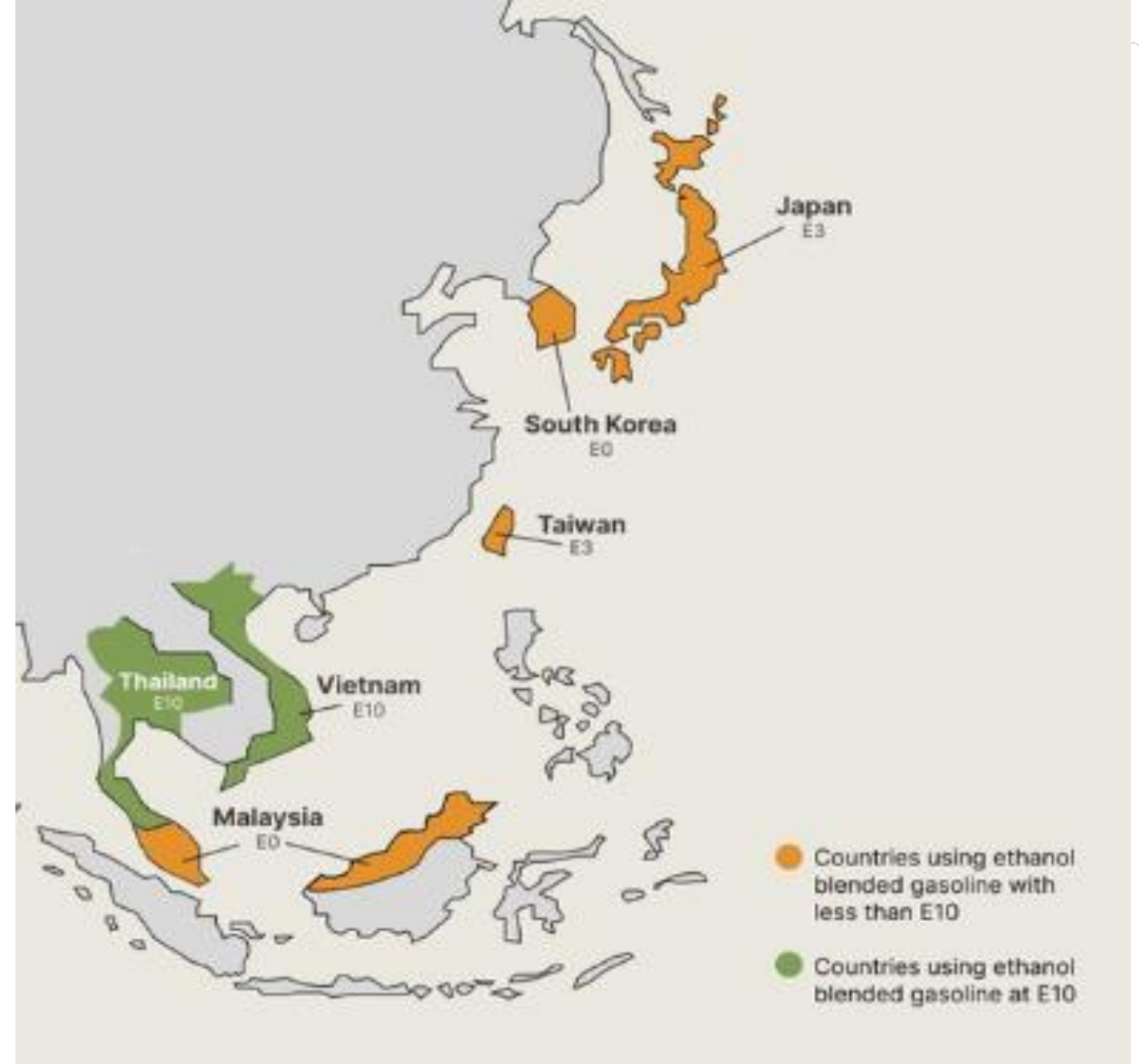


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in South Korea

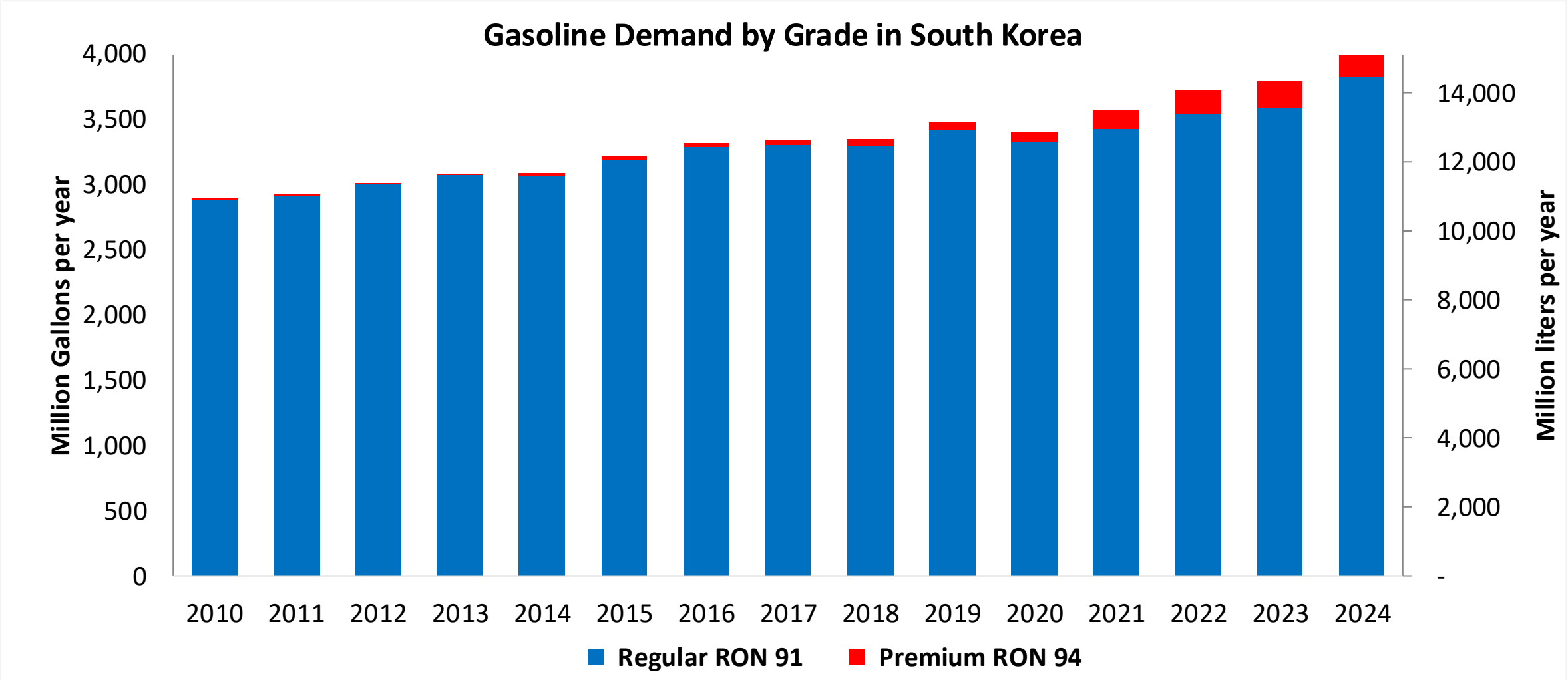


In 2024, gasoline consumption in South Korea was 15,109 million liters (3,991 million gallons) and growth rate was 2.8%. Gasoline production and net imports were 32,661 and -17,552 million liters (8,628 and -4,637 million gallons) correspondingly. South Korea offers two main grades: RON 91 and RON 94) with an average octane of 91.1 RON.

South Korea has implemented requirements for renewable fuels that are currently met with biodiesel blending. Although there is an oxygen requirement, ethanol is not currently blended in gasoline. Efforts to implement pilot programs have been stalled because of a lack of consensus and reluctance of different actors in the petroleum industry.

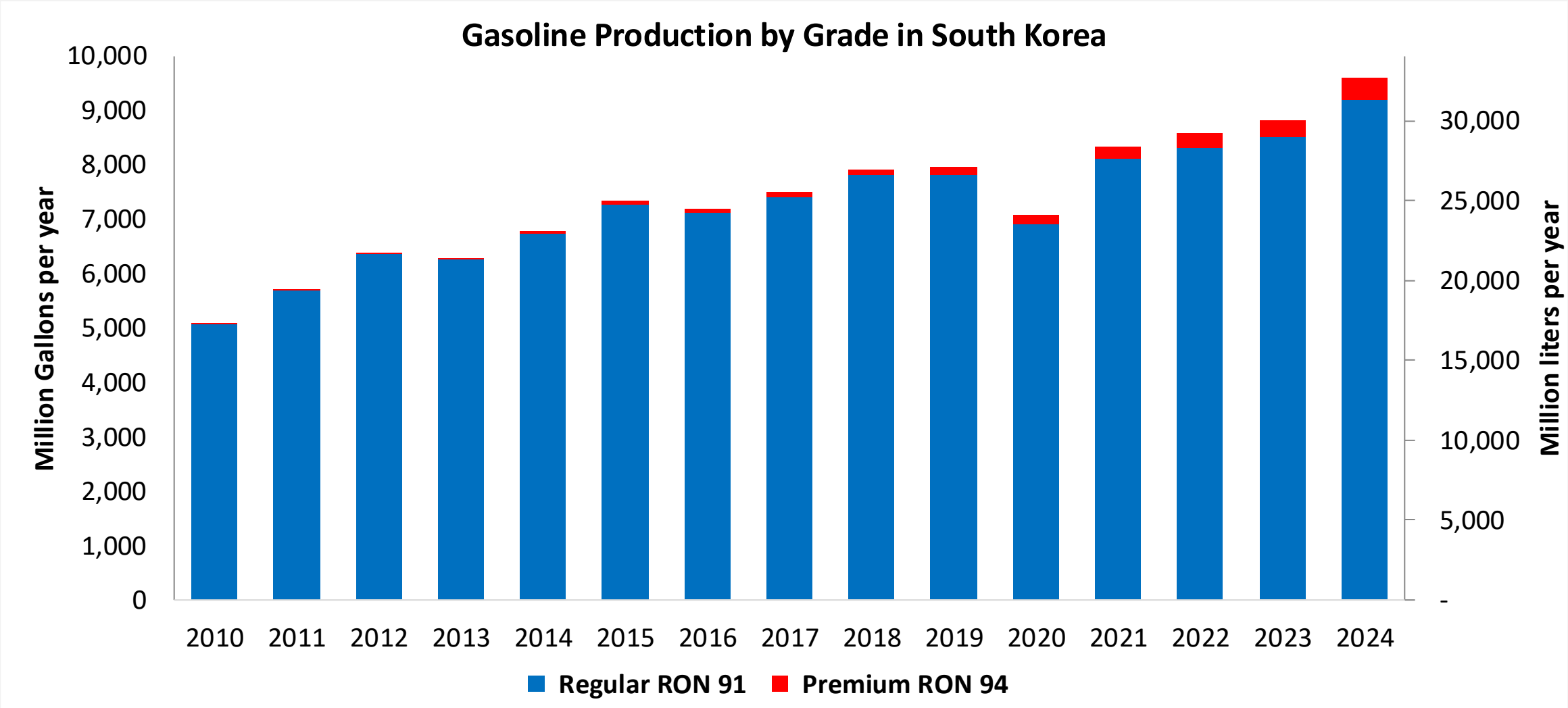
Source: Korean Energy Economics Institute, IEA

Gasoline Demand in South Korea



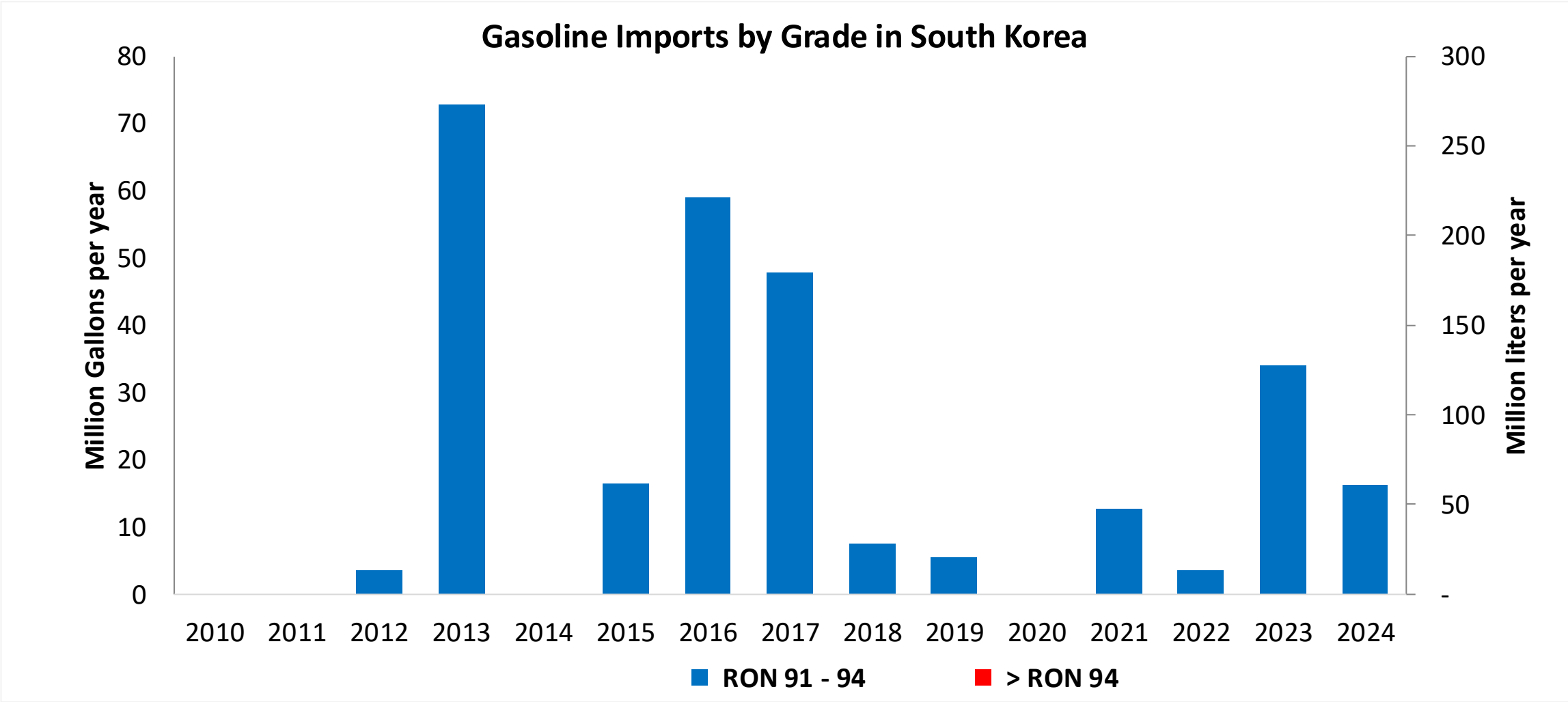
Source: Korean Energy Economics Institute

Gasoline Production in South Korea



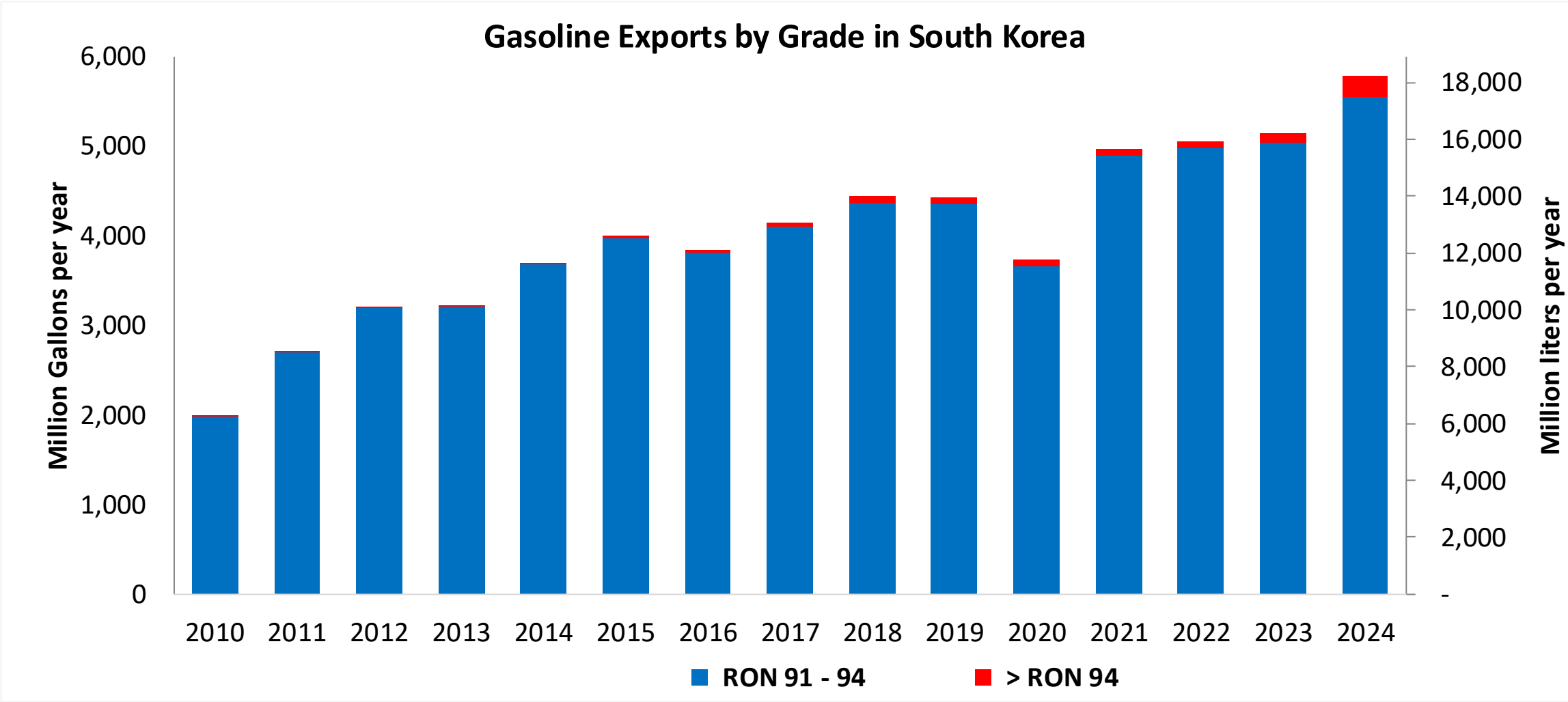
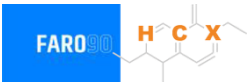
Source: Korean Energy Economics Institute

Gasoline Imports in South Korea



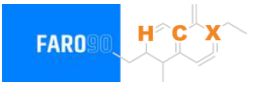
Source: Korean Energy Economics Institute

Gasoline Exports in South Korea



Source: Korean Energy Economics Institute

Ethanol Balance in South Korea

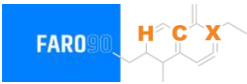


South Korea does not produce or consumes bioethanol in a commercial level for gasoline blending. It imports large quantities of bioethanol for industrial uses.

Gasoline Quality in South Korea

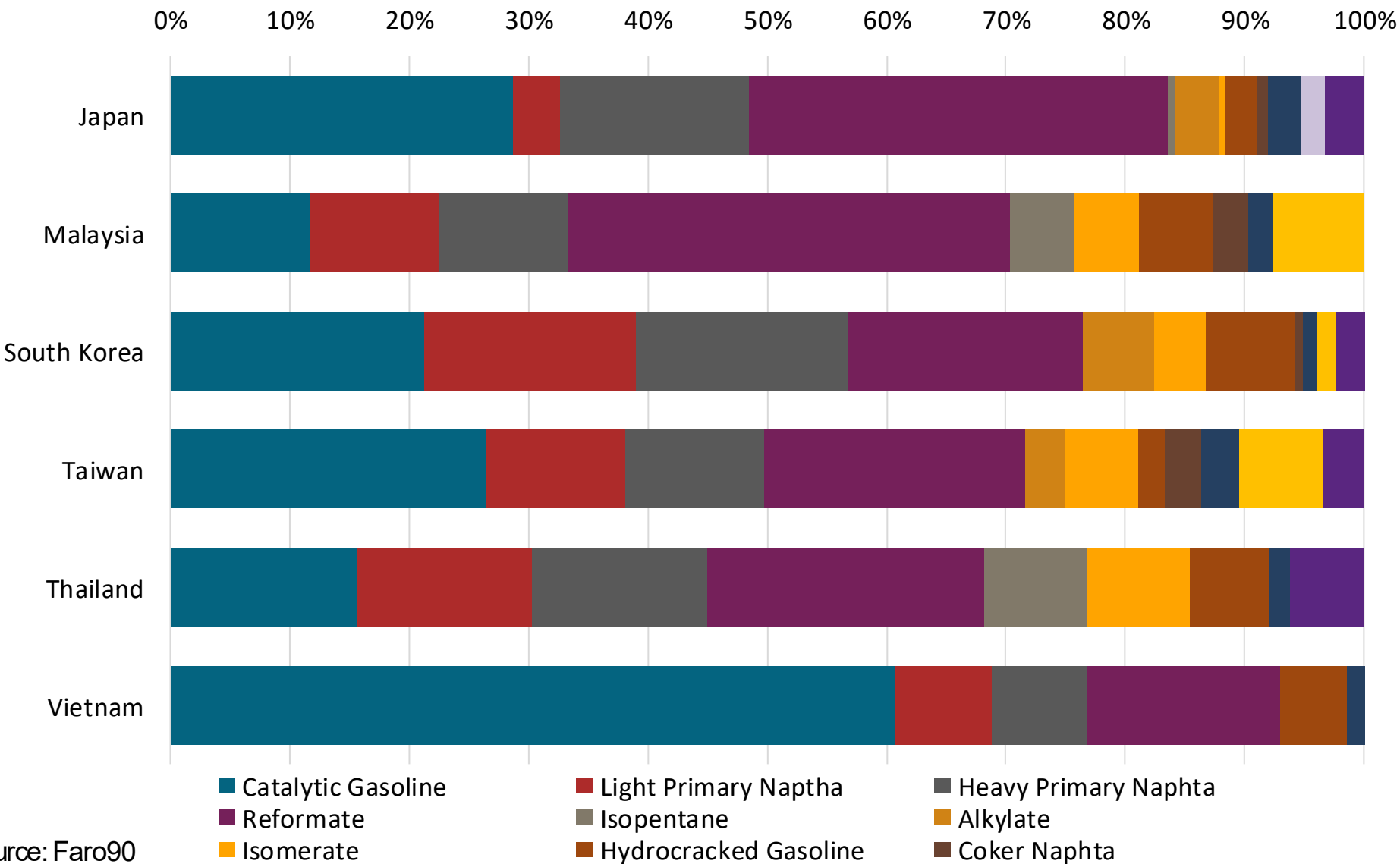
Name	Clean Air Conservation Act			
Implementation Year	2009			
Applicability	Nationwide			
Grade	RON 91		RON 94	
	Min	Max	Min	Max
RON	91.0		94.0	
MON				
AKI				
Benzene (%vol)		0.7		0.7
Aromatics (% vol)		24.0		24.0
Olefins (%vol)		18.0		18.0
Lead (g/L)		0.013		0.013
Manganese (mg/L)		-		-
Sulfur (ppm)		10		10
Oxygen (%v)	0.5	2.3	0.5	2.3
Ethanol (%vol)				
MTBE (%vol)				
RVP (psi)		8.7		8.7

Gasoline Component Blending in Asia



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

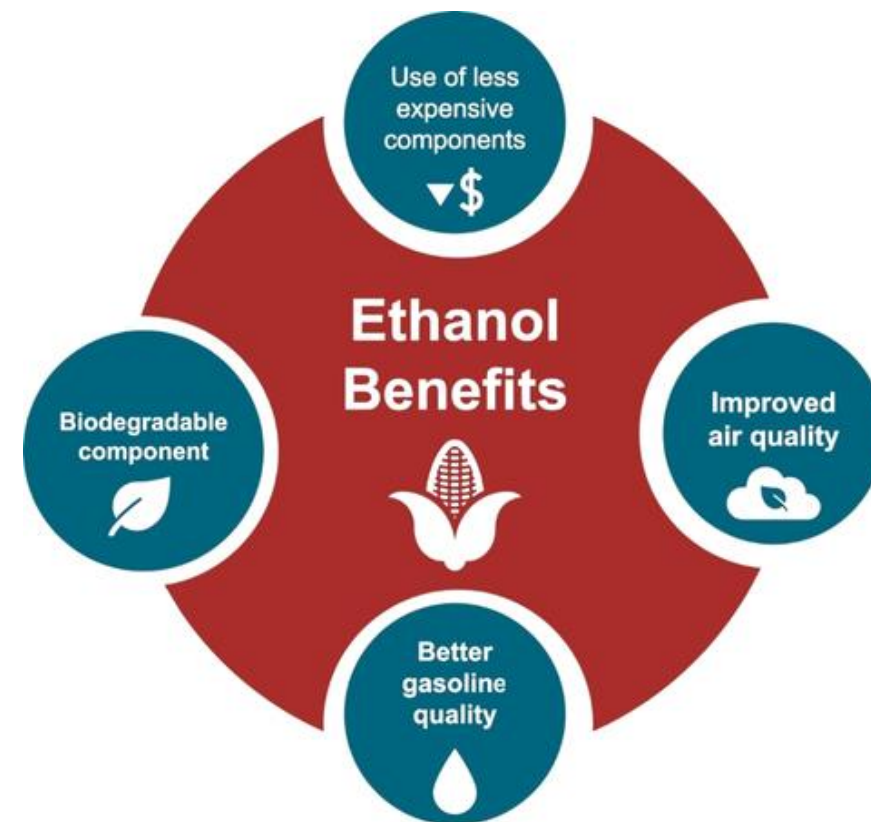
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

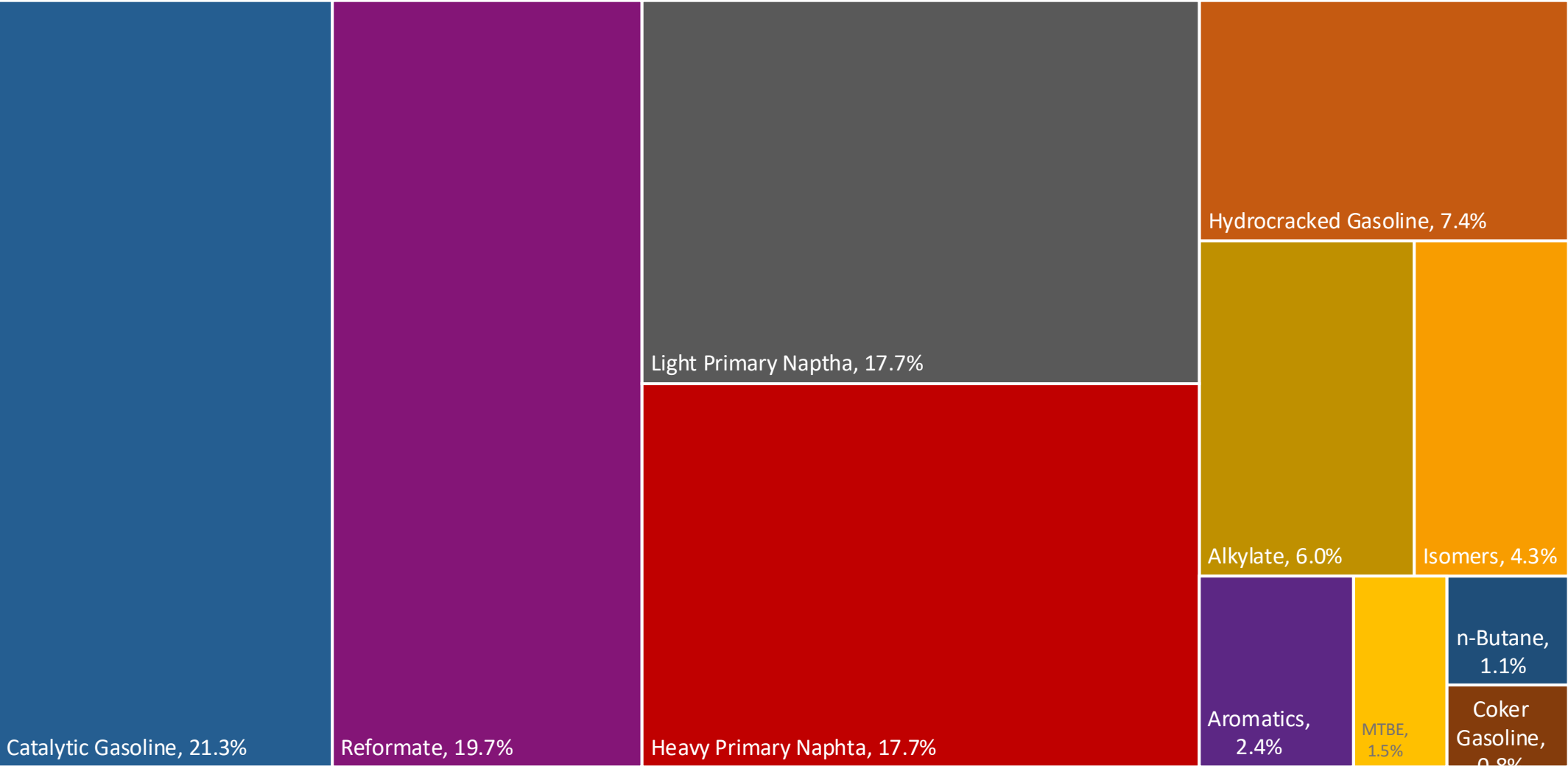
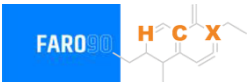
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

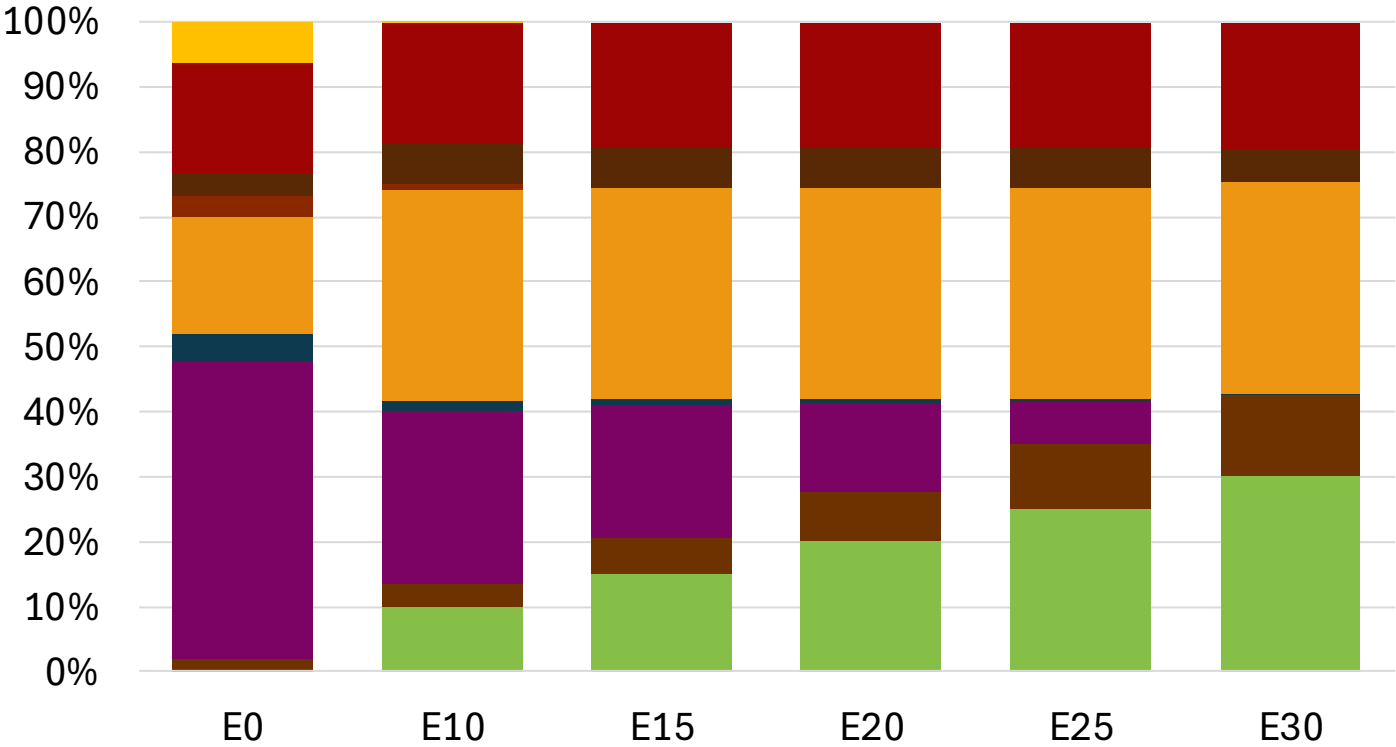
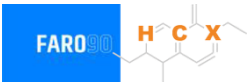


South Korea Available Blending Components



Source: Faro90

South Korea – Gasoline 91 – Constant Octane

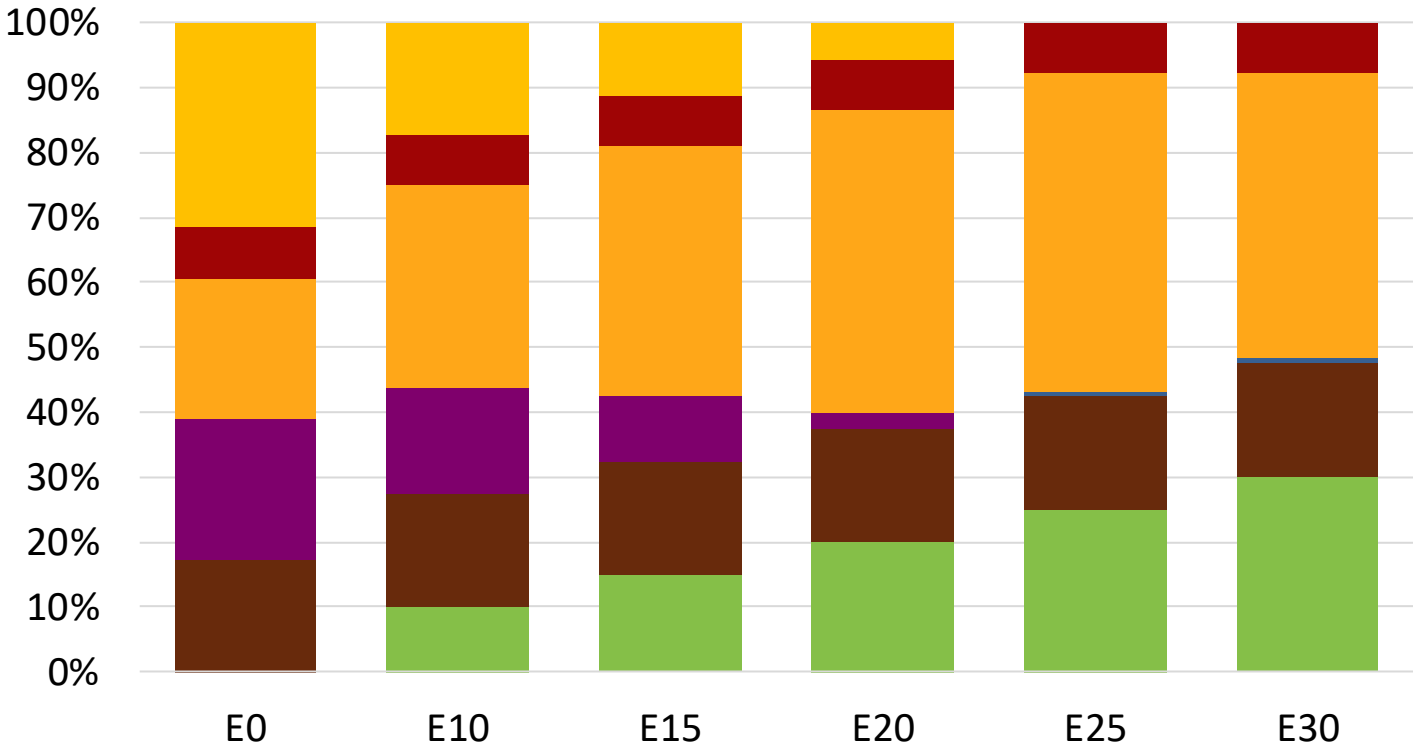
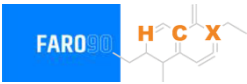


Average CIF 2024
Prices

Octane (RON)	91.0	91.0	92.1	93.2	94.3	95.3
Price (US\$/gal)	\$ 2.257	\$ 2.137	\$ 2.094	\$ 2.050	\$ 2.001	\$ 1.949

Source: Faro90

South Korea - Gasoline 94 - Constant Octane

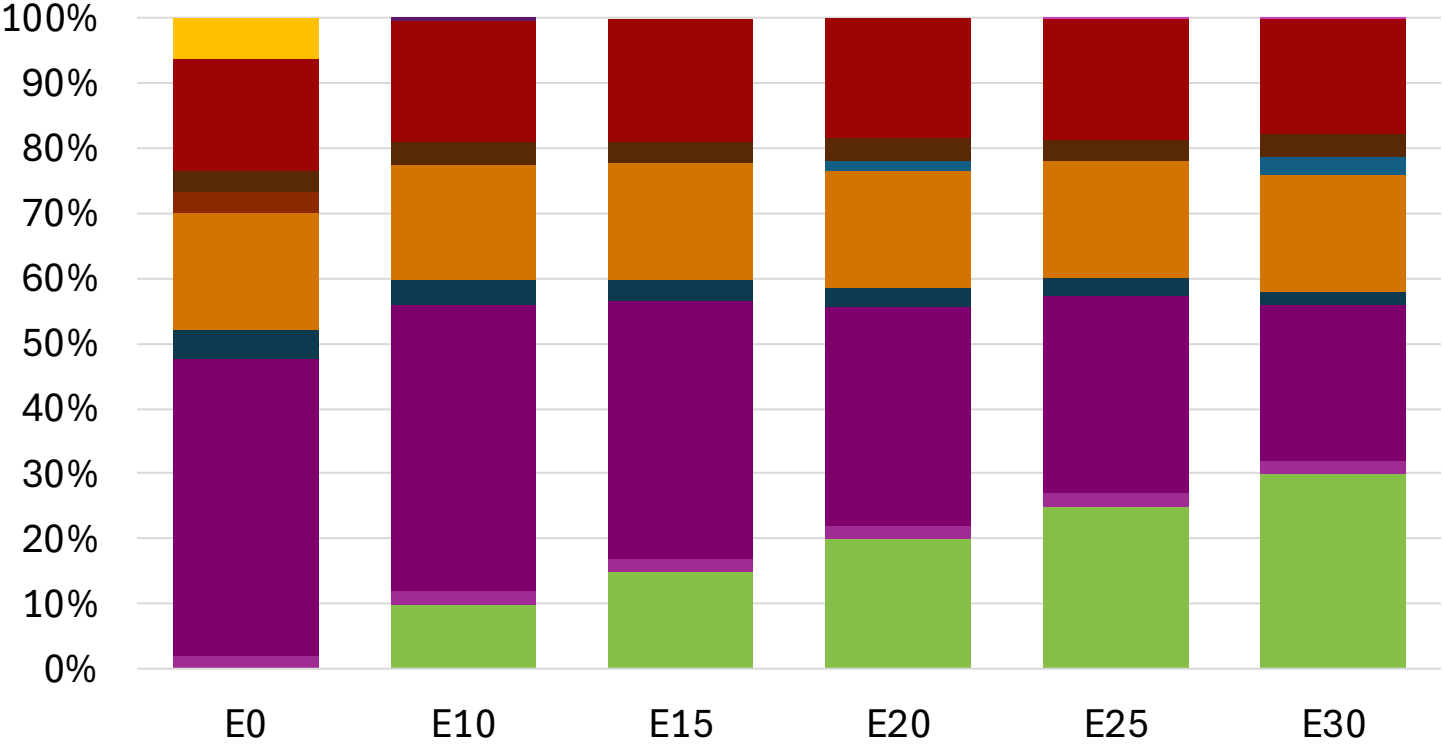


Average CIF 2024 Prices

Octane (RON)	94.0	94.0	94.0	94.0	94.7	96.9
Price (US\$/gal)	\$ 2.240	\$ 2.185	\$ 2.140	\$ 2.086	\$ 2.060	\$ 2.080

Source: Faro90

South Korea – Gasoline 91 – Increased Octane

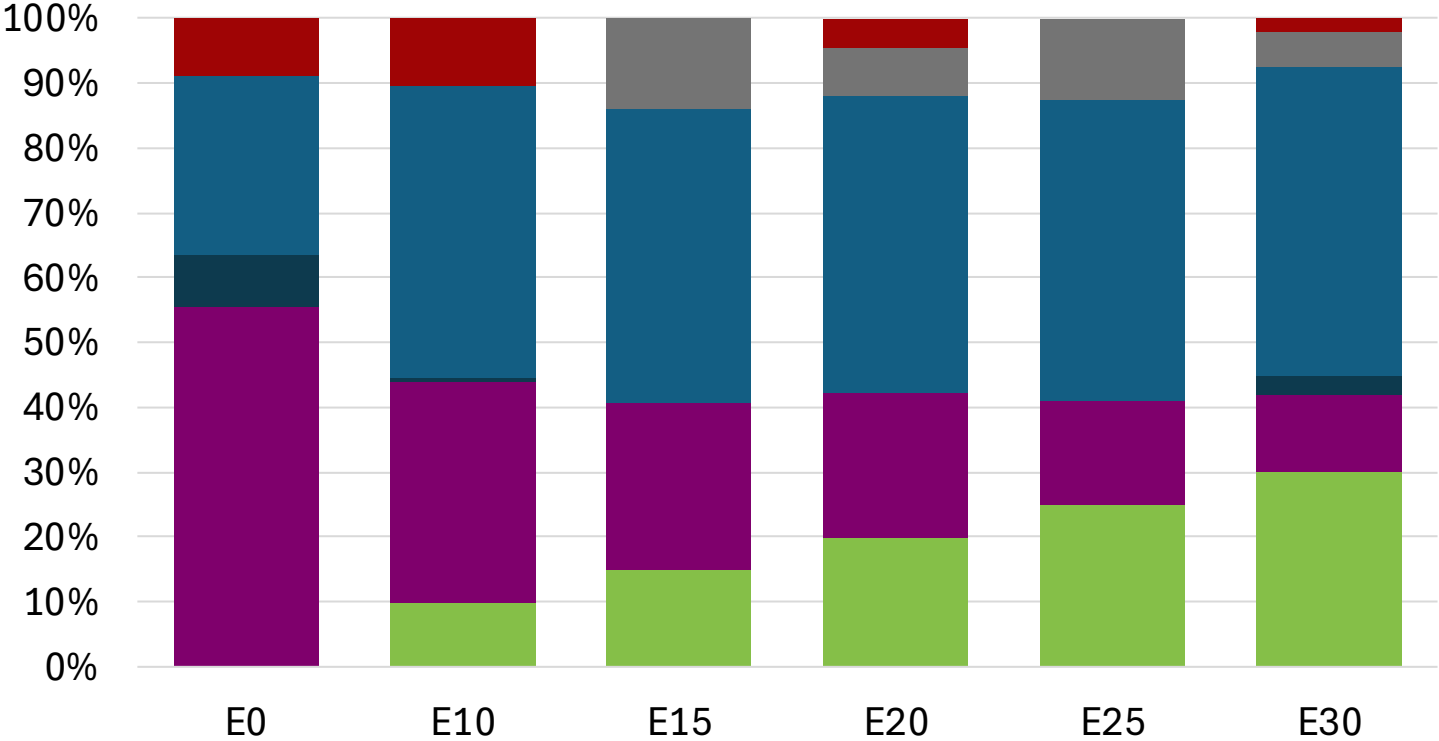
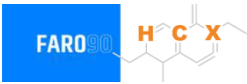


Average CIF 2024
Prices

Octane (RON)		91.0	93.7	95.4	97.1	98.8	100.3
Price (US\$/gal)	\$	2.257	\$ 2.257	\$ 2.257	\$ 2.257	\$ 2.257	\$ 2.257

Source: Faro90

South Korea – Gasoline 94 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	94.0	94.0	94.2	95.6	97.3	99.5
Price (US\$/gal)	\$ 2.240	\$ 2.240	\$ 2.240	\$ 2.240	\$ 2.240	\$ 2.240

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

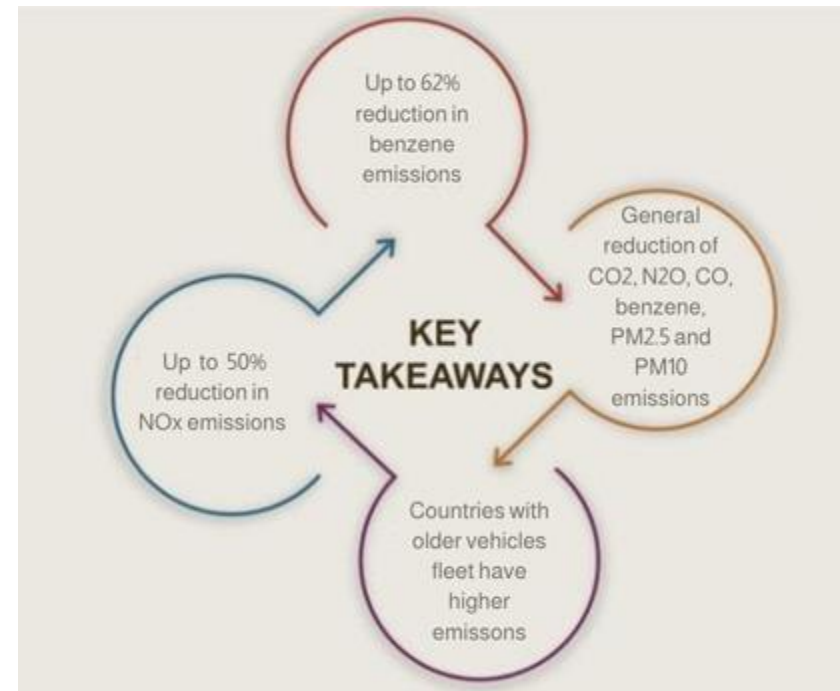
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

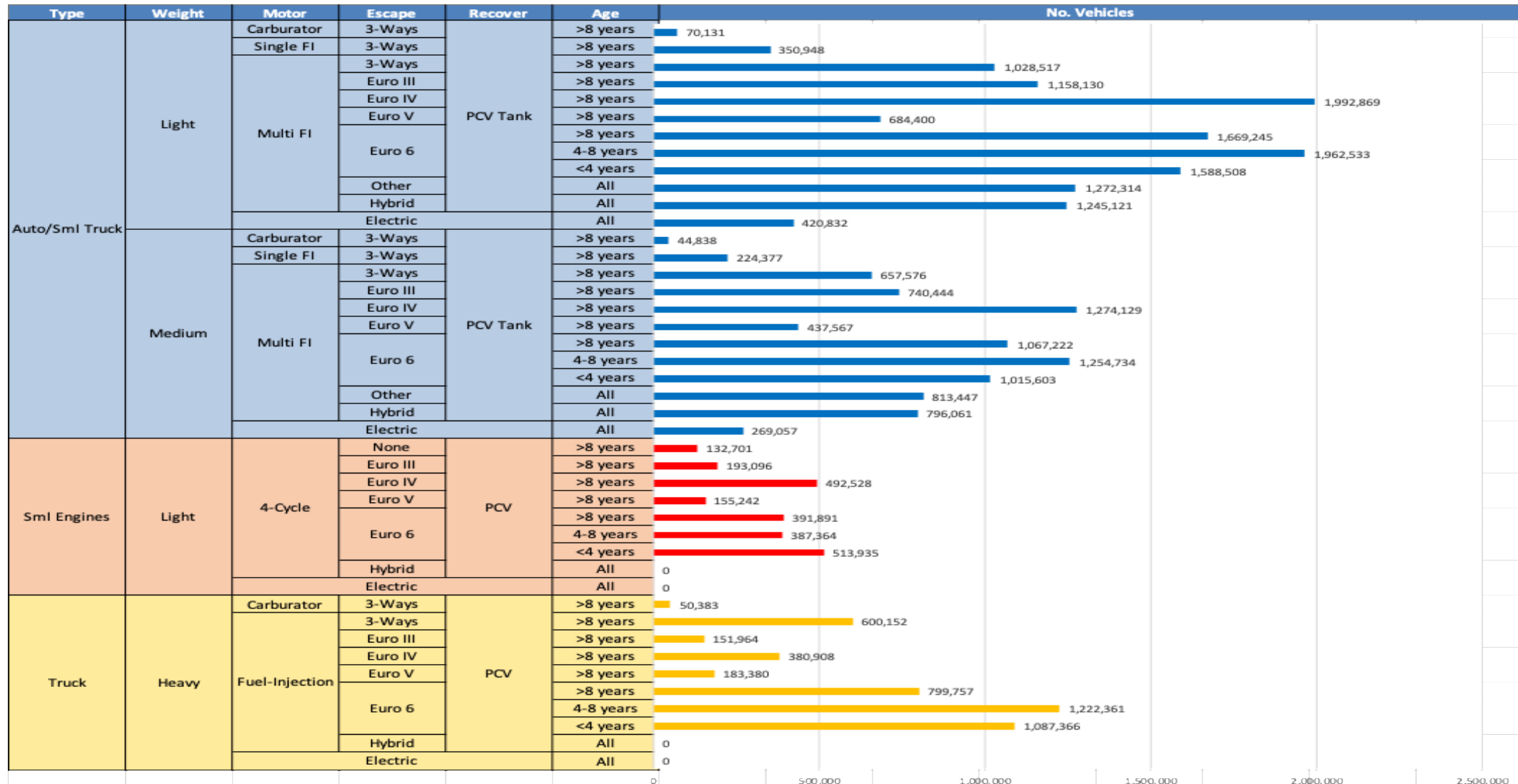
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicle Fleet - South Korea

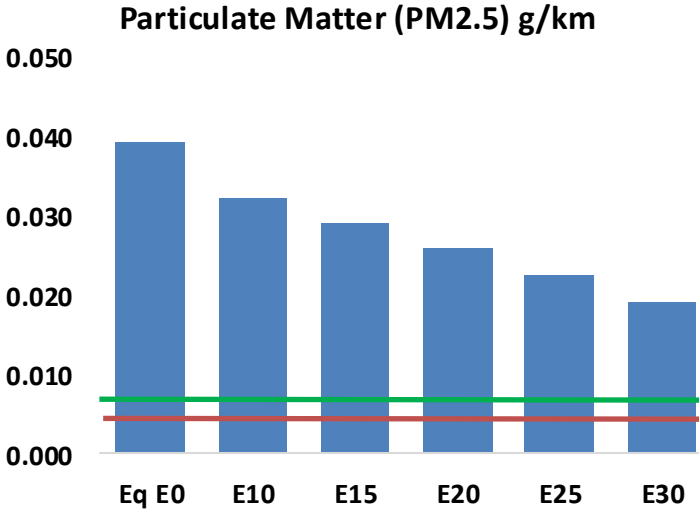
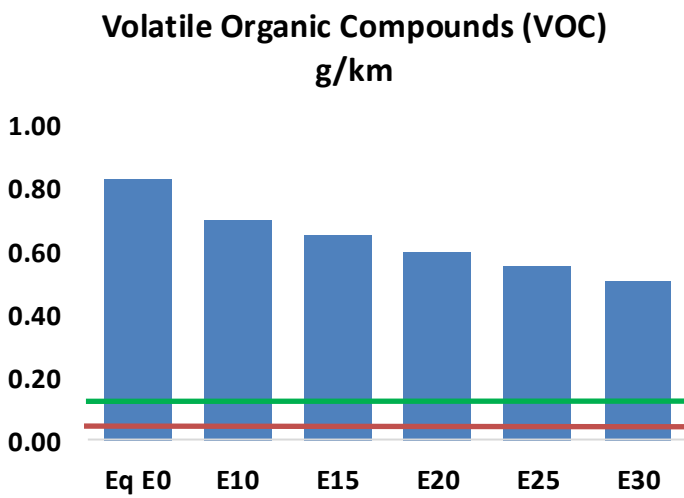
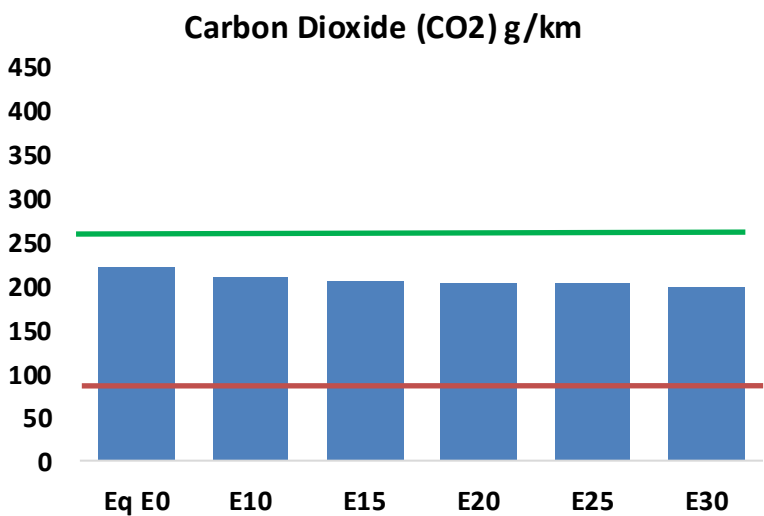
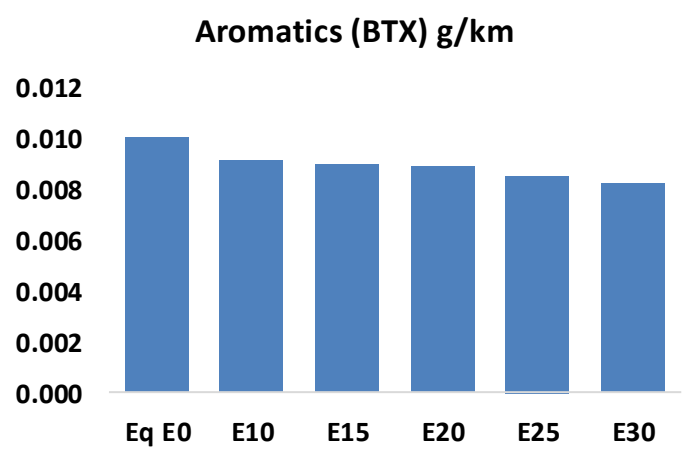
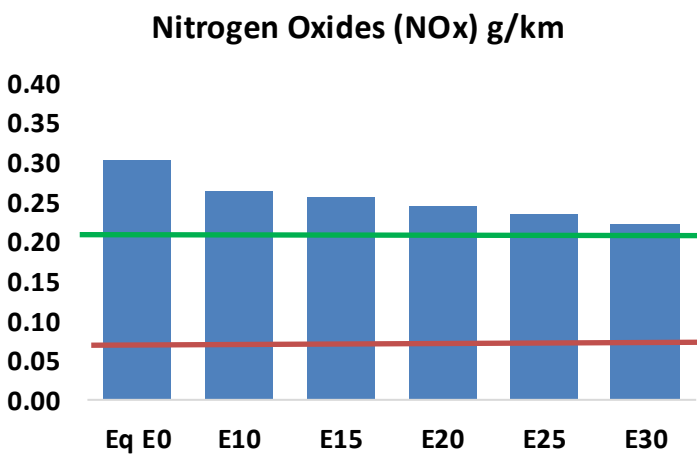
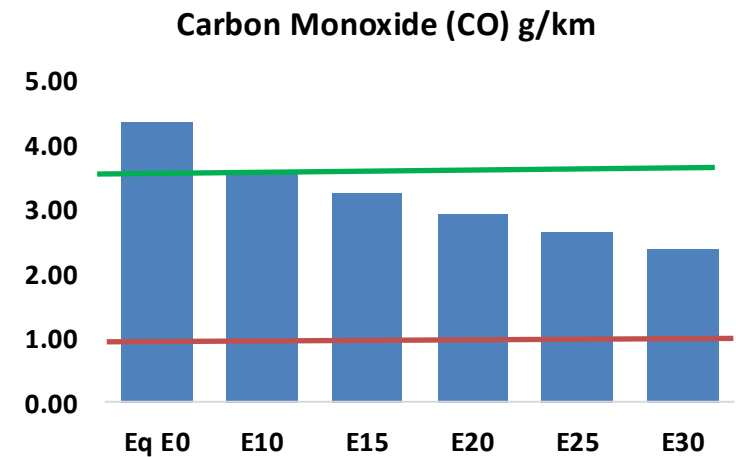
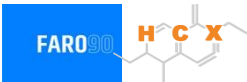


Vehicle Fleet: 28,781,630
Gasoline Vehicle Fleet: 16,830,060

Source: MOLIT

South Korea – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



South Korea – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,591,317	2,359,846	2,128,374	1,932,397	1,732,283	1,577,533	1,414,428	-18%	-33%
CO2	130,279,254	126,929,216	123,765,291	121,276,958	120,045,633	119,459,096	117,257,122	-5%	-8%
VOC	489,471	452,129	414,788	384,244	353,532	329,925	298,394	-15%	-28%
NOx	180,005	166,205	156,882	152,209	144,597	138,669	131,664	-13%	-20%
SOx	1,520	1,405	1,290	1,193	1,100	999	892	-15%	-28%
NH3	46,891	46,427	45,962	46,181	46,700	47,065	47,369	-2%	0%
N2O	2,724	2,709	2,698	2,712	2,768	2,798	2,830	-1%	2%
CH4	101,103	101,103	101,103	102,619	104,844	106,765	108,584	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,005	1,895	1,785	1,704	1,626	1,567	1,484	-11%	-19%
Acetaldehyde	6,582	7,554	11,084	16,879	22,996	26,277	31,049	68%	249%
Formaldehyde	25,164	24,465	28,519	33,488	35,083	38,812	42,251	13%	39%
Benzene	5,947	5,715	5,415	5,330	5,285	5,055	4,891	-9%	-11%
PM2.5	7,324	6,658	5,992	5,406	4,833	4,218	3,575	-18%	-34%
PM10	15,949	14,499	13,049	11,773	10,525	9,186	7,786	-18%	-34%

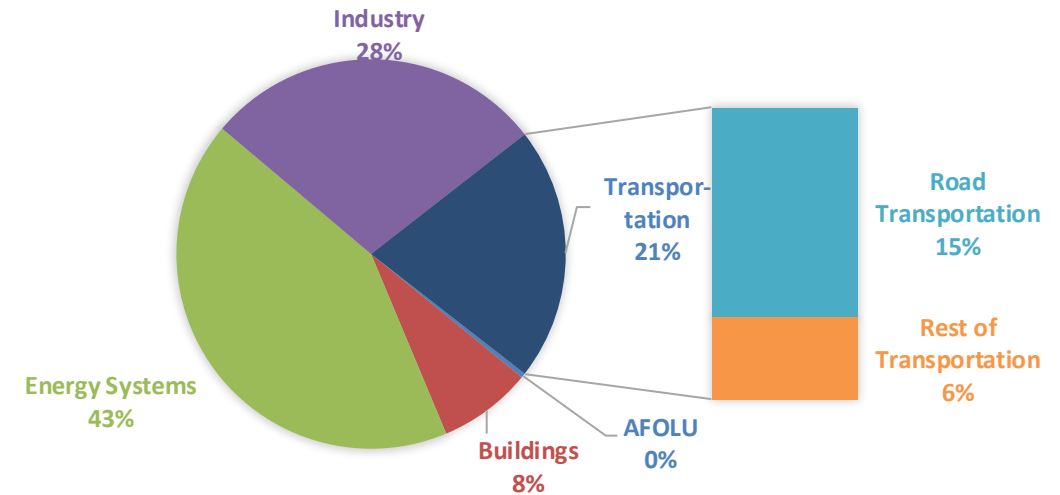
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

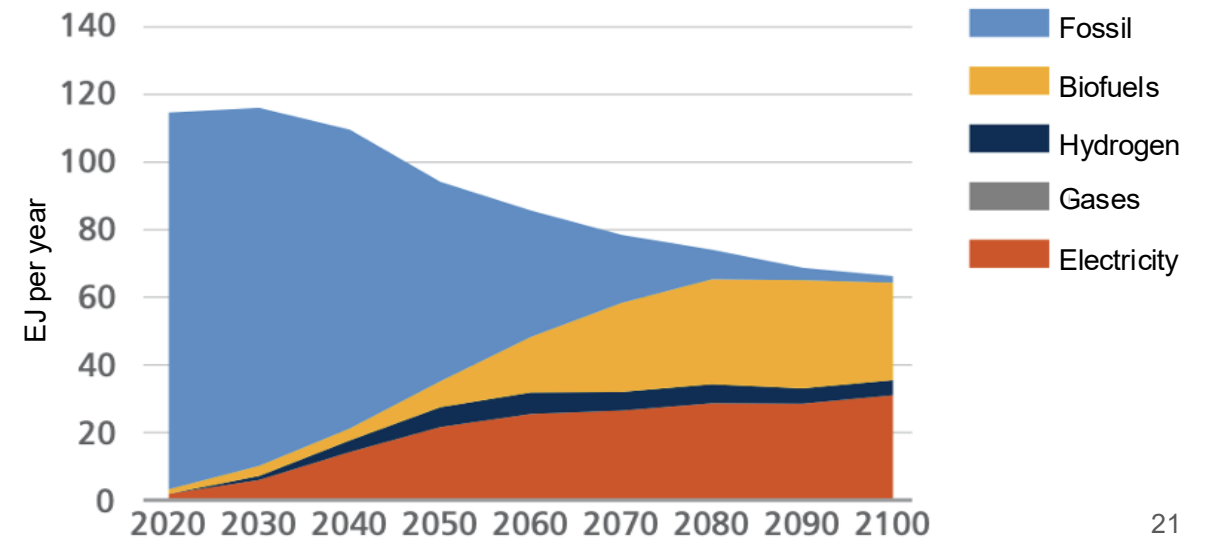
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

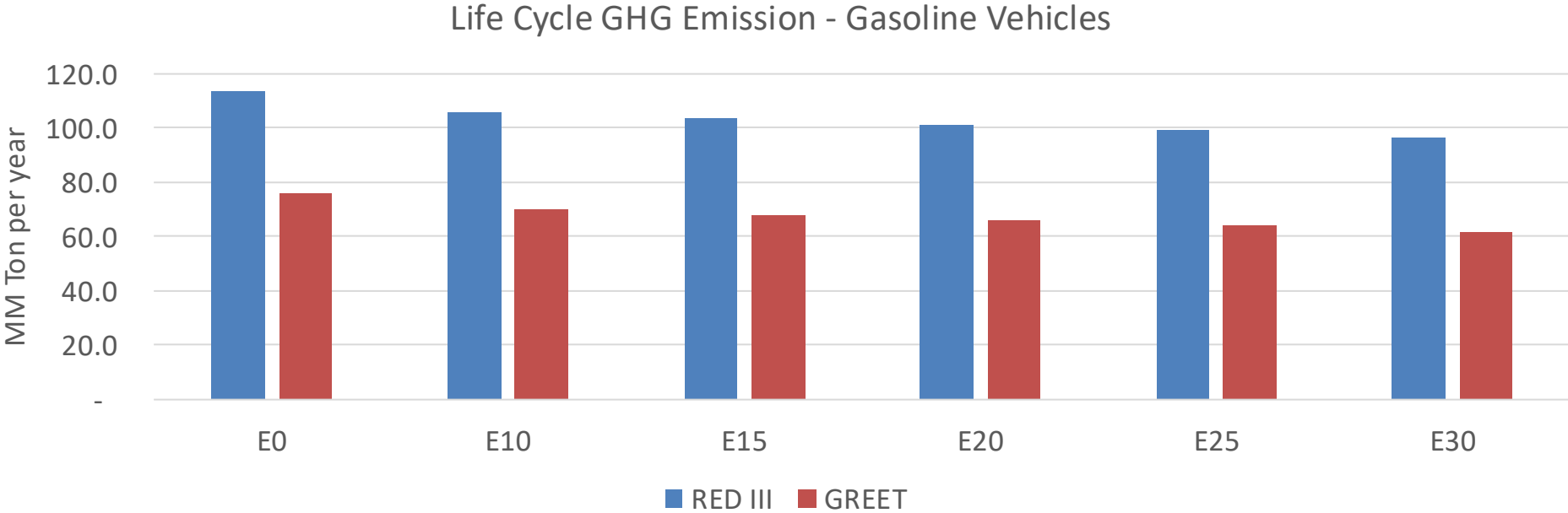
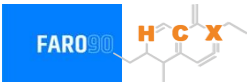
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



South Korea – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	682.3
Target @ 2035 (MMT/year)	332.3
Delta	349.9

vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.1%	13.8%	16.0%	19.1%
RED II/III	0.0%	6.4%	8.7%	10.9%	12.6%	15.1%

Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.8%	2.4%	3.0%	3.5%	4.2%
RED II/III	0.0%	2.1%	2.8%	3.5%	4.1%	4.9%